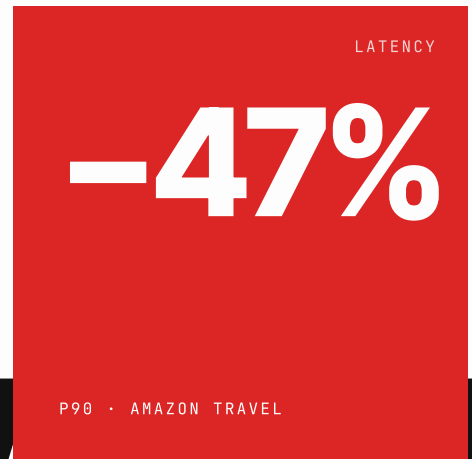


# PRAKHAR SHEKHAR PARTHASARTHI



Senior Software Engineer — Java · Distributed Systems · AWS

Three years at Amazon Travel.

Java microservices on AWS. Hotels, Flights, Insurance.

Ship services. Monitor p90. Keep on-call quiet.

CONV.

**+33%**

ON-CALL

**20+ → 5**

CPT

**-43 bps**

SERVICES

**4 shipped**

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## SDE II / Amazon Travel — Hotels · Flights · Insurance

JUL 2023 — SEP 2025 · BENGALURU

Owned four Java microservices across the funnel. *Latency, resilience, quiet on-call.* Java 17 · Spring · AWS · CDK.

### 01 PARALLELIZED THE HOT PATH.

Refactored serial service fan-out to **CompletableFuture.allOf** with **bulkhead ExecutorService pools** per downstream. p90 fell **1.10s → 0.59s (-47%)**; conversion lifted **+33%**. No new hardware.

### 02 IDEMPOTENT CAPTURE ON SQS + STEP FUNCTIONS.

Dedupe via **DynamoDB idempotency keys** with TTL; dropped double-capture in renewals. CPT **-43 bps**; saved **~30 bps** of funnel-churn across insurance.

### 03 CDK-JAVA REWRITE OF DEPLOYS.

Replaced bespoke Python scripts with typed **CDK-Java** stacks and staged rollouts. Rollbacks **halved**, MTTR down **~1 day**, new services bootstrapped in hours.

### 04 DRAINED THE PAGER.

Rewrote alarms, killed config-hygiene SEVs **5-6 → 0**, on-call pages **20+ → ~5 / shift**. **150+ incidents** authored; stopped **~30 bps** of funnel-churn tied to silent failures.

02

SECTION TWO / RECORD

LEDGER · ARCHIVE · EVIDENCE

Prior roles, stack, projects, education, recognition. Terse. Load-bearing.  
Nothing decorative.

ROLE 02 / SDE I

SDE I — Amazon Travel

JUL 2022 — JUL 2023 · BENGALURU

Shipped REST services with **@Repository DynamoDB + Spring DI**. Owned the auto-insurance renewal path; **+60% renewal completion**. Held unit coverage at **80%+** across JUnit / JaCoCo. Quiet code, loud coverage.

ROLE 03 / INTERN

SDE Intern — Amazon Travel

MAY 2021 — FEB 2022 · BENGALURU

Led the **GraphQL → Java** migration on the renewals surface. Built **One-Click Renewal** end-to-end with **Guice DI**. Coverage held at **90%+**. Converted to full-time on the strength of it.

THE STACK

INSTRUMENTS · EIGHT CATEGORIES

|                                                                                                                              |                                                                                                                                |                                                                                                                          |                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <p><b>JVM</b></p> <p>Java 17 · Spring Boot, MVC, DI · Guice · Maven · Gradle · strict nullability · pattern matching.</p>    | <p><b>CONCURRENCY</b></p> <p>ExecutorService · CompletableFuture · ConcurrentHashMap · ScheduledExecutor · bulkhead pools.</p> | <p><b>AWS</b></p> <p>Lambda · DynamoDB · SQS · Step Functions · API Gateway · AppSync · CloudWatch · ECS · CDK-Java.</p> | <p><b>QUALITY</b></p> <p>JUnit 5 · Mockito · JaCoCo · integration harnesses · contract tests · golden-path fixtures.</p> |
| <p><b>FRONT-END</b></p> <p>React SSR &amp; CSR · Redux · TypeScript · Vitest · accessible forms · edge-rendered bundles.</p> | <p><b>DATA PATTERNS</b></p> <p>DynamoDB single-table · idempotency keys with TTL · dedupe queues · outbox writes.</p>          | <p><b>OPS</b></p> <p>Runbooks · alarm hygiene · SEV pages · on-call rotations · graceful degradation.</p>                | <p><b>METHOD</b></p> <p>Trunk-based · code-review discipline · blameless postmortems · SLO-first design.</p>             |

PROJECTS

OUT OF HOURS · EVIDENCE

Redis Server · in Java

[github.com/praxstack/redis-server-java](https://github.com/praxstack/redis-server-java)

Single-binary Redis-wire implementation — SET / GET / INCR / EXPIRE over RESP, event loop, AOF persistence. Benchmarks on commodity laptop; ~78% of C-Redis on my box.

|              |              |           |          |
|--------------|--------------|-----------|----------|
| SET 154K RPS | GET 195K RPS | INCR 128K | 42 TESTS |
|--------------|--------------|-----------|----------|

Markdown Viewer App

[github.com/praxstack/markdown-viewer-app](https://github.com/praxstack/markdown-viewer-app)

Tiny Markdown reader. **Zero runtime dependencies**. Bundle <100 KB gzipped. Held by **155+ Vitest tests**; CI gate won't ship below threshold.

|            |            |           |
|------------|------------|-----------|
| 155+ TESTS | <100 KB GZ | ZERO DEPS |
|------------|------------|-----------|

EDUCATION · TRAINING

- M.C.A. — Computer Science**  
Motilal Nehru National Institute of Technology, Prayagraj  
2019 — 2022 · GPA 9.25 / 10
- B.Sc. (Hons) — Computer Science**  
Bhaskaracharya College of Applied Sciences, Univ. of Delhi  
2015 — 2018

RECOGNITION · CITATIONS

- Amazon Pay — Merchants Categories Champion 2023**. Awarded for owning the renewals funnel across the merchant insurance line.
- 100+ trust-buster bugs** closed across travel quality sprints — contributed to **~30 bps of funnel-churn** stopped at source.